Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: QUANTUM COMPUTING – 1(24CS3207)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	Noise-Robust Quantum Support Vector Machines for Classification	
S. No.	**University ID**	**Name**
1	2420030544	Alaparthy Mohnish
2	2420030265	Pedireddi Sandeep
3	2420030199	Koppada Leela Krishna Likith Raj
4		
Name of the Guide		MS. SIRIPURI DIVYA

Abstract

1. **Problem:** Classical support vector machines (SVMs) rely on kernel functions to separate data that is not linearly separable, but certain complex, high-dimensional data distributions remain computationally expensive for classical kernels to model efficiently. Quantum Support Vector Machines (QSVMs) address this by using quantum circuits to compute kernel similarities in exponentially large Hilbert spaces. However, near-term quantum devices (NISQ-era) are inherently noisy, and it remains unclear how much this noise degrades real-world classification performance — a gap that limits practical deployment of QSVMs.

2. **Why quantum computing:** Quantum feature maps encode classical data into quantum states, allowing similarity measures (kernels) that are conjectured to be classically hard to reproduce for certain data structures. This offers a theoretical path to enhanced expressivity over classical kernels, motivating renewed interest in quantum-enhanced machine learning as quantum hardware access becomes more widely available.

3. **Existing solutions/projects**: Havlíček et al. (2019) introduced the foundational QSVM framework using quantum kernel estimation with the ZZFeatureMap, demonstrating quantum-kernel separability on synthetic benchmark datasets. Subsequent work has explored alternative feature maps (e.g., PauliFeatureMap) and applications to small classical datasets. However, most existing studies evaluate QSVMs primarily under ideal, noise-free simulation, leaving their robustness under realistic hardware noise conditions comparatively underexplored.

4. **Solution**: This work implements and benchmarks QSVM classifiers using multiple quantum feature maps (ZZFeatureMap and PauliFeatureMap) on standard classification datasets, first under ideal simulation and then under Qiskit Aer noise models approximating real NISQ device conditions. Classification accuracy, kernel matrix fidelity, and decision boundary quality are systematically compared against classical SVM baselines across increasing noise levels and circuit depths.

5. **Expected results**: We expect classical SVMs to maintain stable accuracy across all conditions, while QSVM performance is expected to degrade progressively with increasing noise and circuit depth, with the degradation rate varying by feature map choice. The study aims to quantify this noise sensitivity and identify which feature map configurations offer the most practical robustness, providing empirical guidance for deploying QSVMs on near-term quantum hardware.which are thought to be hard to mimic using classical methods for certain types of data.

Signature of the Guide HOD